TUNIS BUSINESS SCHOOL
UNIVERSITY OF TUNIS

University of Tunis
Tunis Business School

Fall 2022

Production & Operation Management BA 305
Midterm Exam
(Duration 1.5 hours)

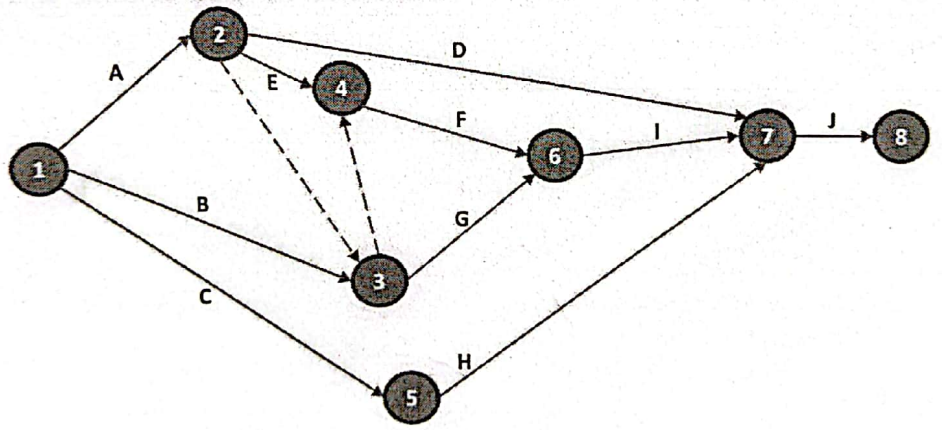
Question 1

A manufacturer of shopping baskets for supermarkets has recently purchased new equipment allowing it to reduce the labor time required to produce a basket. Initially, the company used 5 employees producing together an average of 80 baskets per hour. The cost of labor was 10 dinar per employee per hour and the cost of machines is 40 dinar per hour. With the new production system, one of the employees was transferred to another department. In this case, the total production increases by 4 baskets per hour, while the cost of equipment increases by 10 dinar.

1. Compare the two production systems using the number of baskets produced per employee per hour as a measure.
2. Compare the two production systems according to their overall productivity
3. Comment on the variations in productivity according to the two measures and identify the most relevant for this situation.

Question 2

Consider the following graph of a given project:



The table below provides all necessary information:

Activity	Regular time (weeks)	Crash time (weeks)	Normal cost (TND)	Crash cost (TND)
A	6	5	2500	2950
B	4	4	3000	-
C	7	5	5500	6200
D	6	6	3800	-
E	4	3	4800	5100
F	5	5	3900	-
G	6	4	5400	6800
H	4	4	3500	-
I	5	5	3000	-
J	4	3	4000	4600

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1. Identify the project duration at normal time
2. Identify the project cost at normal time
3. The project manager wants to reduce the project duration as much as possible. He has a budget of 850 TND for project crashing. How should he operate and what would be the new project duration and cost?

Question 3

Scrap of low-quality products of AB company for the last 6 days of production is given as follows:

Period	Scrap
1	2000
2	2200
3	2360
4	2500
5	2200
6	2400

1. Use exponential smoothing with α=0.1 to forecast for period 7 given that the forecast for the first week has been estimated to be 1800.
2. Repeat exponential smoothing with α =0.2 and use whatever reliable tool to provide a judgement on which value of α is more likely to be suitable.
3. Consider a two-period moving average. Provide the forecast for the seventh period and state which among the obtained forecasts seems to be more reliable with the necessary justifications.

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